TopStep ICT + Microstructure Trading Bot

Technical Documentation & Honest Findings

Futures bot for TopStep via the ProjectX/TopstepX API — ICT signal engine, deterministic risk management, ML position sizing, deep-learning research pipeline, and live order-flow capture.

1. Executive summary

This system trades CME index/crypto micro futures (MNQ, MES, MBT) on a TopStep funded/eval account. It is built so that every trading idea is **validated before it risks money**: signals are rule-based and transparent, the risk engine caps downside to the eval fee, and an ML layer scales position size by a model-estimated win probability — but only a model that has passed out-of-sample (walk-forward) validation is ever allowed to size live.

Honest note: The headline research result so far is that the ICT bar-pattern strategy has **no out-of-sample directional edge** (win rates land on the random-walk baseline). The system correctly refused to deploy it. Current work tests whether true order-flow / microstructure data carries an edge that 1-minute bars cannot. No profitable model has been shipped; this document describes the machinery and the evidence, not a proven money printer.

2. System architecture

The live loop, identical in paper (sim) and live mode — only the broker differs:

```
feed (quotes/trades/DOM)  ->  1-min bars + tape capture
  ->  ICT signal engine  ->  setup?
  ->  gates (R:R, killzone, flat, risk)
  ->  ML confidence  ->  size = risk_cap x confidence_scale
  ->  bracketed order (server-side OCO stop+target)
  ->  manage: trailing-DD floor watch, flatten before close
  ->  learn: label each trade win/loss -> retrain
```

- **Broker layer** — abstract interface; *ProjectXBroker* (live REST + SignalR) and *SimBroker* (paper fills against the live feed) are interchangeable.
- **Data layer** — 1-min bar aggregation, multi-contract history backfill, and DuckDB capture of bars + the tape (every trade with aggressor side) + book snapshots.
- **Signal engine** — ICT primitives fused into risk-defined setups.
- **Risk engine** — models TopStep rules; the hard ceiling on size.
- **ML stack** — confidence/sizing model, deep research models, walk-forward validation.
- **Discord control** — status, pause/resume, flatten, manual-approve, live alerts.

3. Data layer

- **Live feed:** ProjectX SignalR hubs stream quotes, the trade tape (with buy/sell aggressor), and DOM. Auth is an API-key login (24h JWT).

- **History:** /History/retrieveBars gives OHLC only; we stitch several contract months into ~2-3 months of 1-min bars per instrument for backtesting.
- **Capture-forward:** because no tick history is sold to us, the bot records the live tape and book into DuckDB, building the microstructure dataset over time.
- **Settlement/labels:** trades are walked forward bar-by-bar (stop checked before target = conservative) and booked net of the round-turn fee.

4. ICT signal engine

ICT concepts are defined **operationally** (testable, not discretionary):

- **Market structure:** fractal swing highs/lows; BOS (break of structure = trend continuation) and CHoCH (change of character = reversal); displacement = an impulsive bar whose body dwarfs ATR.
- **Fair Value Gaps (FVG):** 3-candle price imbalances; tracked until mitigated.
- **Order Blocks:** the last opposing candle before a displacement that breaks structure.
- **Liquidity pools & sweeps:** clustered equal highs/lows where stops rest; a sweep is a wick through the pool that closes back inside (stop-grab + rejection).
- **Sessions / killzones:** Asian range, London & NY killzones (ET); entries only inside killzones.

The two setup archetypes

- **Reversal** ('price reverts to a point'): a liquidity sweep + rejection, confirmed by a CHoCH or displacement; stop beyond the swept wick, target the next opposing liquidity pool.
- **Continuation** ('momentum / bull-bear switch'): an established trend that printed a BOS, entered on a retrace into an unmitigated FVG/order block; stop beyond the zone, target the next draw on liquidity.

How stop and target are chosen

Stop = the structural invalidation level (the swept wick, or the far edge of the FVG/order block) plus a 0.25*ATR buffer — if price trades there, the idea is wrong. **Target** = the nearest opposing liquidity pool, optionally floored to a minimum reward:risk multiple. R:R = (target-entry)/(entry-stop).

Honest note: R:R is a **planned, geometric** ratio — it does not include win probability or fees. A high win rate at small targets can still be net-negative once fees are paid. This is exactly what the data showed.

5. Risk management (the actual edge mechanism)

Most funded accounts fail on the drawdown rules, not on signal. The risk engine models TopStep's End-of-Day **trailing** max drawdown, the daily-loss lock, max contracts, a loss-streak halt, and a give-back lock — and enforces them on **real-time intraday equity**, flattening the instant equity touches the trailing floor.

Account	Profit target	Trailing DD	Daily loss	Max contracts
50K	$3,000	$2,000	$1,000	5
100K	$6,000	$3,000	$2,000	10

150K	$9,000	$4,500	$3,000	15

Sizing = deterministic cap x ML confidence. The risk engine computes the hard ceiling (contracts such that a full stop ~= per-trade risk, also bounded by distance-to-floor and the account max). The ML confidence (0-1) can only scale this **down**. Confidence never increases risk past the ceiling — that is the safety contract. Sizing amplifies an edge; it cannot create one (N contracts of a negative-EV trade lose N times faster).

6. The AI / ML stack

- **Confidence model** — outputs P(win) for a setup. Cold start = a transparent hand-weighted logistic *prior* (not learned alpha); upgrades to a trained scikit-learn logistic once >=200 labeled trades exist.
- **Deep model** — a 1D-CNN over the recent price-pattern window fused with the engineered feature vector (PyTorch, MPS-accelerated), heavily regularized for the modest dataset.
- **Directional model** — drops the ICT rules; predicts triple-barrier direction from the price window + microstructure features, trading only when confident.
- **Walk-forward validation** — train on the past, test on data never seen, roll forward; pool out-of-sample trades and bootstrap a confidence interval on mean PnL. A model ships only if OOS expectancy > 0 with the CI above zero. This is the line that separates a real edge from a curve-fit.

7. Results so far — the honest verdicts

Three independent, rigorous out-of-sample tests, all on ~143k bars / months of data:

Test	OOS result	Verdict
ICT setups + CNN selector	win 47%->66%, but -$2.41/trade (CI below 0)	no edge
Timeframe x R-multiple sweep (6 configs)	every config -EV; win rate = random baseline	no edge
Pure-ML direction (price only)	OOS accuracy ~44-49.5% (coin flip)	no edge

Interpretation: the CNN **works** as a selector (it lifted win rate to 66% and scored AUC ~0.74), but the underlying trades don't clear fees, and at fixed R-multiples the win rates fall exactly on the driftless random-walk baseline (~1/(1+R)). Conclusion: **1-minute bar patterns carry no directional information** on this data — consistent with market efficiency at this horizon and with prior research in this repo.

8. The microstructure pivot (current work)

Profitable discretionary day traders read the **order book and the tape** — order-flow imbalance, absorption, sweeps, book pressure — none of which survive compression into a 1-min OHLC bar. So the tests above ruled out bar-pattern edge, NOT microstructure edge. The bot now captures the real data:

- **Tape capture** — every trade with its aggressor side, into DuckDB (working; ~tens of thousands of trades/session during RTH).
- **Order-flow features** — signed-volume imbalance, VPIN-style toxicity, aggressor-run persistence, large-print ratio, trade intensity, book pressure / micro-price.
- **Model** — a sequence CNN over micro-buckets of the tape + these features, predicting short-horizon triple-barrier direction, judged by the same walk-forward gate.

- **Book/DOM capture** — subscribed but not yet populating (entitlement/parse follow-up).

> **Honest note:** A live parsing bug had been silently discarding the entire feed (quotes key the contract as 'contract'; trades arrive as a list) — fixed, so capture now works. The microstructure model has not yet been validated; it will be tested OOS once enough tape accumulates, and shipped only if it clears the same gate. It may also turn out to be ~random — that is an honest possible outcome.

9. How to run & check progress

```
# paper-trade the live feed + capture order flow (zero account risk)
python -m topstep_bot.run --mode sim --no-discord

# check live progress (reads a heartbeat, no DB lock)
python -m topstep_bot.status

# backtest the ICT strategy on captured/stitched history
python -m topstep_bot.backtest --instrument MNQ

# build + walk-forward the microstructure model when enough tape exists
python -m topstep_bot.ml.microstructure --instrument MNQ

# continuous deep-learning loop: re-evaluate as data grows, ship only if +EV OOS
python -m topstep_bot.ml.deep_train --loop --refresh-days 10
```

10. Honest limitations & next steps

- No profitable model has been validated or shipped. The bot trades on a conservative prior and is intended for sim until a model clears OOS.
- Modest data (~2-3 months bars; tick data only from now). Deep learning is data-hungry; results are small-sample until more accumulates.
- Live SignalR payloads, OCO bracket behavior, and DOM capture need confirmation on a live account before real size.
- Sim fills are optimistic (touch fills, 1-tick slippage); real fills will be worse.
- Next: accumulate RTH tape, run the microstructure walk-forward, and either wire it live (if +EV OOS) or report honestly that intraday edge is not capturable at retail latency.

This document reflects the system and evidence as built. Negative results are reported as negative; nothing here is presented as a guaranteed edge.